

Perdiction Results

2024-10-02

Data Summary:

- **Number of observations for bill passage predictions:** 39,996. 4 requests returned null responses.
- **Number of observations for bill summary completion:** 40,000. A total of 53 responses exceeded the token limit of 750, and 1 response was dropped due to content filtering.

Results

Bill Passage

Table 1: True Bill Passage Rates

Chamber	True Passage Rate
Pass House	0.078
Pass Senate	0.062

Table 2: Performance Evaluation of LLM Predictions of Bill Passage

Predicted Outcome	Model	Prompt	Accuracy	TPR	FPR
Pass House	GPT-3.5	w/o date restriction	0.223	0.976	0.840
Pass House	GPT-3.5	w/ date restriction	0.230	0.979	0.833
Pass House	GPT-4o	w/o date restriction	0.914	0.183	0.024
Pass House	GPT-4o	w/ date restriction	0.711	0.626	0.282
Pass Senate	GPT-3.5	w/o date restriction	0.322	0.937	0.719
Pass Senate	GPT-3.5	w/ date restriction	0.315	0.951	0.727
Pass Senate	GPT-4o	w/o date restriction	0.929	0.217	0.024
Pass Senate	GPT-4o	w/ date restriction	0.760	0.676	0.234

Bill Summary Completion

Table 3: Performance Evaluation of LLM Completion of Bill Summaries (Non-Embedded)

Model	Prompt	Average
GPT-3.5	w/o date restriction	0.000
GPT-3.5	w/ date restriction	0.001
GPT-4o	w/o date restriction	0.000
GPT-4o	w/ date restriction	0.000

Table 4: Performance Evaluation of LLM Completion of Bill Summaries (Embedded)

Metric	Model	Prompt	Average	Baseline	Relative to Baseline ($\uparrow$ Similar)
Cosine Similarity	GPT-3.5	w/o date restriction	0.495	0.271	0.825
Cosine Similarity	GPT-3.5	w/ date restriction	0.504	0.270	0.869
Cosine Similarity	GPT-4o	w/o date restriction	0.505	0.266	0.900
Cosine Similarity	GPT-4o	w/ date restriction	0.523	0.271	0.930
Euclidean Distance	GPT-3.5	w/o date restriction	0.994	1.205	0.175
Euclidean Distance	GPT-3.5	w/ date restriction	0.985	1.207	0.184
Euclidean Distance	GPT-4o	w/o date restriction	0.983	1.210	0.188
Euclidean Distance	GPT-4o	w/ date restriction	0.964	1.205	0.200